Individual Assignment 5

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1. Set Up

Packages

```
# general use
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# reading .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# making plots
library(patchwork)
library(NatParksPalettes)
```

Data

```r
# taxonomic info
taxon_list <- read_csv(here("data","taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects"
)

# site info
sites <- read_xlsx(
   here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
   sheet = "sites"
)

# water quality
water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+")
)
```

2. Cleaning

`sites` object

```r
# creating new object
ncos_sites <- sites |>
  # filtering to only include ncos quarterly sampling sites
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

clean `taxon_list`

```r
taxon_list_clean <- taxon_list |>
  # convert taxon names to snakecase
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))
```

clean `water_quality`

```r
water_quality_clean <- water_quality |>
  clean_names() |>
  semi_join(ncos_sites, by = "site") |>
  unite("date_site", date, site, remove = FALSE) |>
  group_by(date, site, date_site) |>
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  ungroup() |>
  filter(date_site != "2022-08-12_NVBR")
```

clean `aq_ins`

```r
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
```

```
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter (sum abundance divided by 7.5
   ↪  liters)
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
  left_join(
  select(water_quality_clean,
         date_site,
         med_ph,
         med_sal,
         med_do),
  by = "date_site"
) |>
  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # add full names of sites
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
   ↪  = TRUE),
         site_full = fct_rev(site_full))
```

clean `aq_ins_clean`

```
# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
```

```r
  # filter to only include copepods
  filter(taxon == "copepod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
```

Problems

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

- x-axis: site
- y-axis: ostracod abundance
- geometry to represent average abundance/L: `geom_jitter`
- geometry to represent medians: `stat_summary()`

b. Wrangle your data

```r
# create new object from aq_ins_clean
ost_abundance <- aq_ins_clean |>
  # filter to only ostracods
  filter(taxon == "ostracod") |>
  # remove missing abundance/L values
  filter(!is.na(ave_lit)) |>
  # group by site
  group_by(site_full) |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # ungroup data frame
  ungroup()

slice_sample(ost_abundance, n = 10)
```

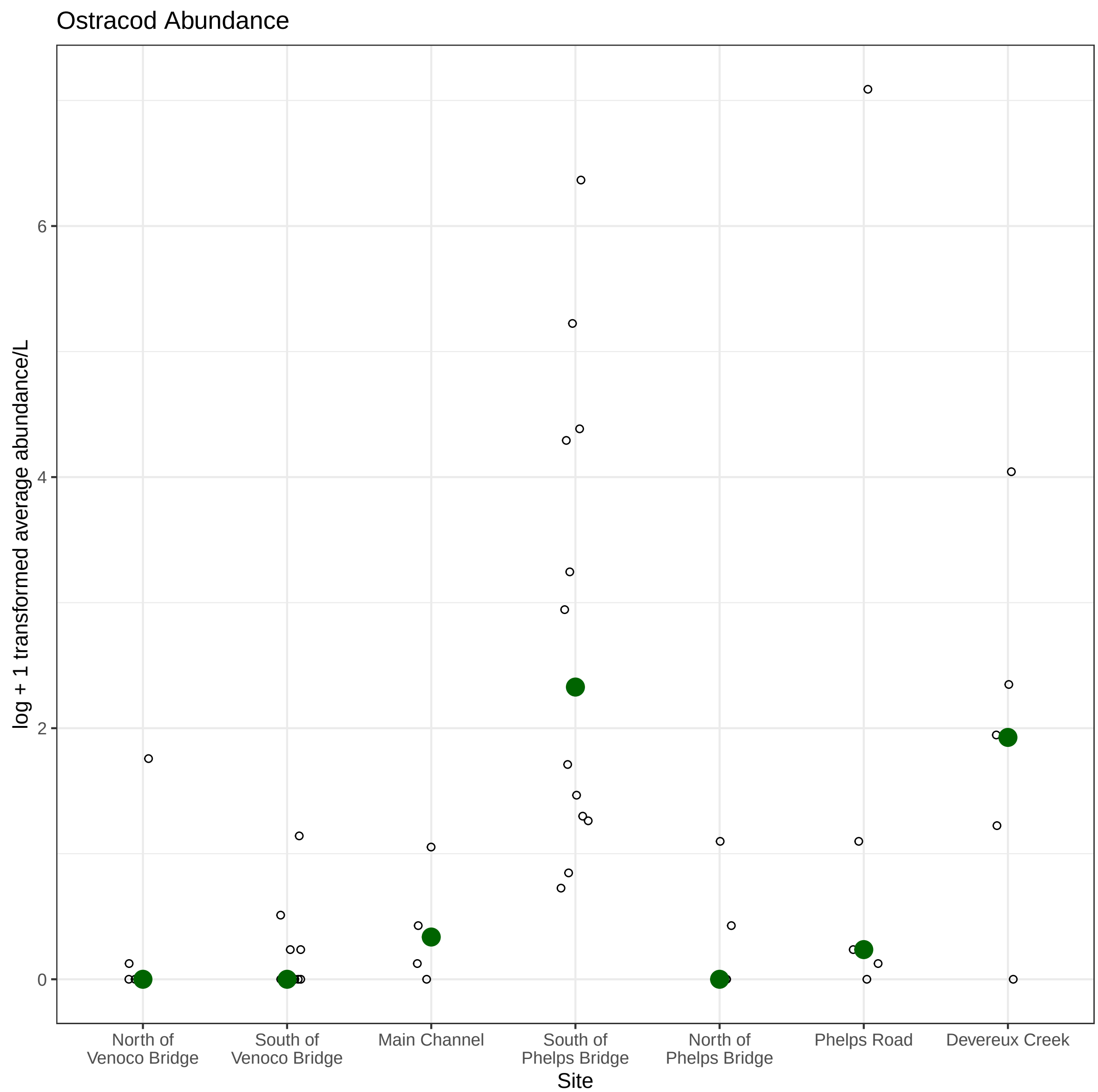
```
# A tibble: 10 x 24
   date_site       date                site  taxon ave_lit med_ph med_sal med_do
   <chr>           <dttm>              <chr> <chr>   <dbl>  <dbl>   <dbl>  <dbl>
 1 2023-11-12_NMC  2023-11-12 00:00:00 NMC   ostr~   0.533 NA        47     14.0
 2 2020-02-21_NVBR 2020-02-21 00:00:00 NVBR  ostr~   4.8     7.34     4.1    6.1
```

```
 3 2023-11-14_NPB  2023-11-14 00:00:00 NPB   ostr~  18       NA       NA      NA
 4 2022-11-19_NDC  2022-11-19 00:00:00 NDC   ostr~   6       NA       NA      NA
 5 2022-01-21_NEC  2022-01-21 00:00:00 NEC   ostr~   0       NA       NA      NA
 6 2022-02-25_NEC  2022-02-25 00:00:00 NEC   ostr~   0        8.75    44.1     9.87
 7 2022-05-06_NMC  2022-05-06 00:00:00 NMC   ostr~   0        9.96     1.45   16.5
 8 2023-02-21_NEC  2023-02-21 00:00:00 NEC   ostr~   0        8.56    NA       8.8
 9 2021-11-16_NPB  2021-11-16 00:00:00 NPB   ostr~  72.1      7.3      2.38   12.1
10 2022-11-12_NEC  2022-11-12 00:00:00 NEC   ostr~   0        7.33     7.01    2.4
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code your figure

```
# base layer
ostracods_plot <- ggplot(data = ost_abundance,
                   # x-axis: site
                   # y-axis: mean abundance/L
                   mapping = aes(x = site_full,
                                 y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               color = "darkgreen") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Ostracod Abundance") +
  # different theme
  theme_bw()

ostracods_plot
```



d. Create a multipanel figure

```
# base layer
salinity_plot <- ggplot(data = aq_ins_clean,
       # x-axis: site
       # y-axis: mean salinity
       mapping = aes(x = site_full,
```

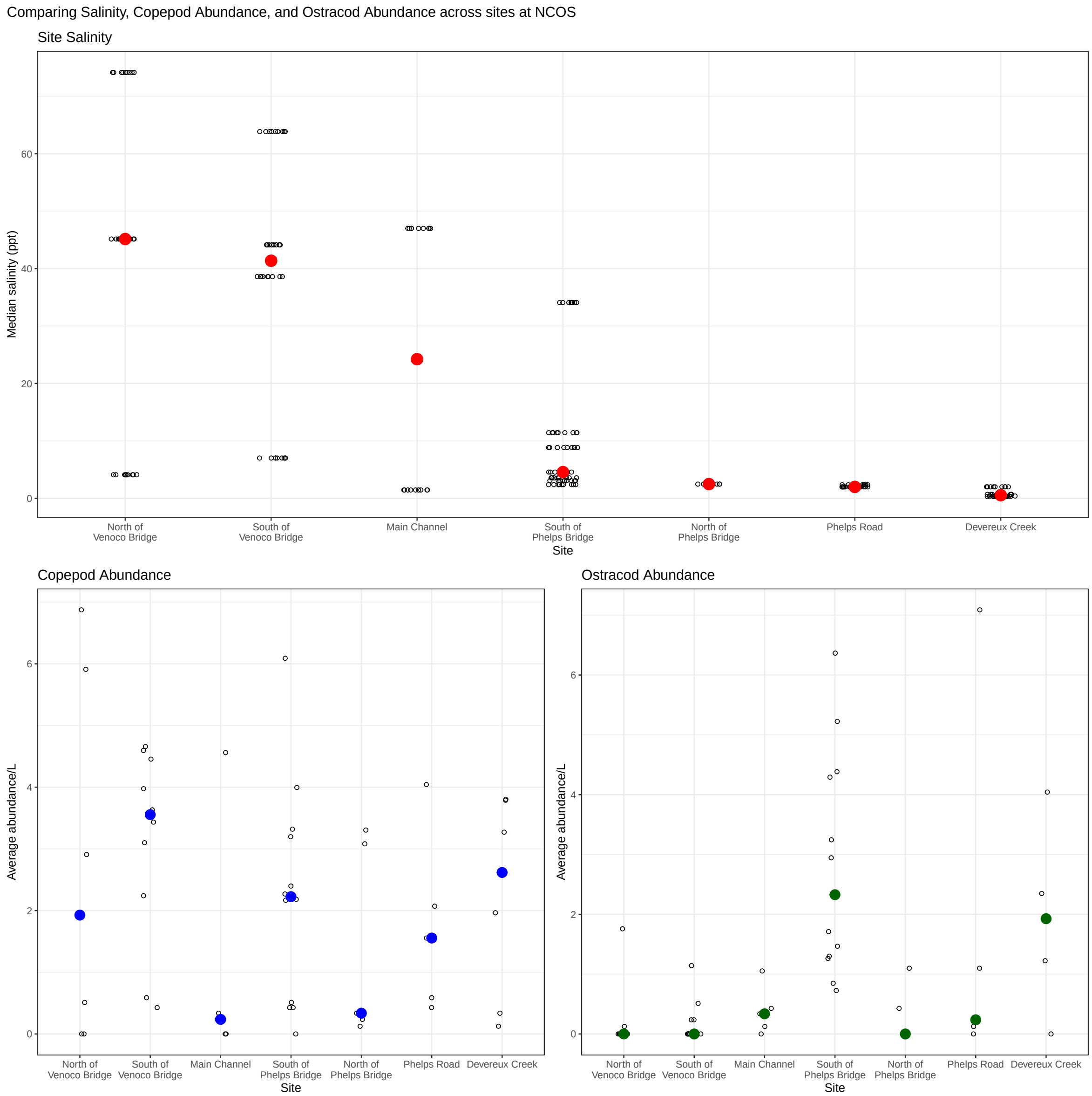
```
                    y = med_sal)) +
  # first layer: salinity measurements at each survey
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # second layer: points to represent means
  stat_summary(fun = median,
               color = "red",
               size = 1) +
  # relabelling x- and y-axes
  labs(x = "Site",
       y = "Median salinity (ppt)",
       title = "Site Salinity") +
  # different theme
  theme_bw()

# base layer
copepods_plot <- ggplot(data = copepods,
                   # x-axis: site
                   # y-axis: mean abundance/L
                   mapping = aes(x = site_full,
                                 y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               color = "blue") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "Average abundance/L",
       title = "Copepod Abundance") +
  # different theme
  theme_bw()

# base layer
```

```
ostracods_plot <- ggplot(data = ost_abundance,
                   # x-axis: site
                   # y-axis: mean abundance/L
                   mapping = aes(x = site_full,
                                 y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               color = "darkgreen") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "Average abundance/L",
       title = "Ostracod Abundance") +
  # different theme
  theme_bw()

(salinity_plot) / (copepods_plot + ostracods_plot) +
  plot_annotation(title = "Comparing Salinity, Copepod Abundance, and
   ↪  Ostracod Abundance across sites at NCOS")
```



e. Interpret your figure

South of Phelps Bridge and Devereux Creek had the highest Average abundance/L measurements across sites for Ostracods. The other sites had pretty consistent lower average abundance/L measurements. North of Venoco Bridge and South of Phelps Bridge had the highest Average abundance/L measurements across sites for Copepods. Main Channel and North of Phelps Bridge had the lowest average abundance/L measurements for Copepods. I could determine these comparisons by observing how high up on the y axis data points lie, and compare across sites on the x axis.

There does not appear to be a strong correlation between site salinity and ostracod abundance. The site with the highest average salinity (ppt) measurements, South of Vencoco Bridge, does not have a notably different average abundance/L measurements than the other sites. Similarly, the site with the highest average abundance/L measurements, South of Phelps Bridge, did not have comparibly high of low salinity (ppt) points along the y axis.

2. Visualizing total abundance as a function of salinity

a. Plan your figure

- x-axis: med_sal
- y-axis: log_ave_lit
- color: taxon
- geometry: geom_point()
- facets: facet_wrap(~ taxon, scales = "free")

b. Wrangle your data

```
# create new object from aq_ins_clean
abundance_salinity <- aq_ins_clean |>
  # create log-transformed abundance column
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # remove underscores from taxon names
  mutate(taxon = str_to_sentence(str_replace_all(taxon, "_", " "))) |>
  # group by taxon
  group_by(taxon) |>
  # calculate max abundance
  mutate(max_abundance = max(log_ave_lit, na.rm = TRUE)) |>
  # un group data frame
  ungroup() |>
  # reorder taxa by maximum abundance
  mutate(taxon = fct_reorder(taxon,
                             max_abundance,
                             .desc = TRUE)) |>
  # keep needed columns
  select(taxon,
         med_sal,
         log_ave_lit)
```

```r
# display 10 random rows
abundance_salinity |>
  slice_sample(n = 10)
```

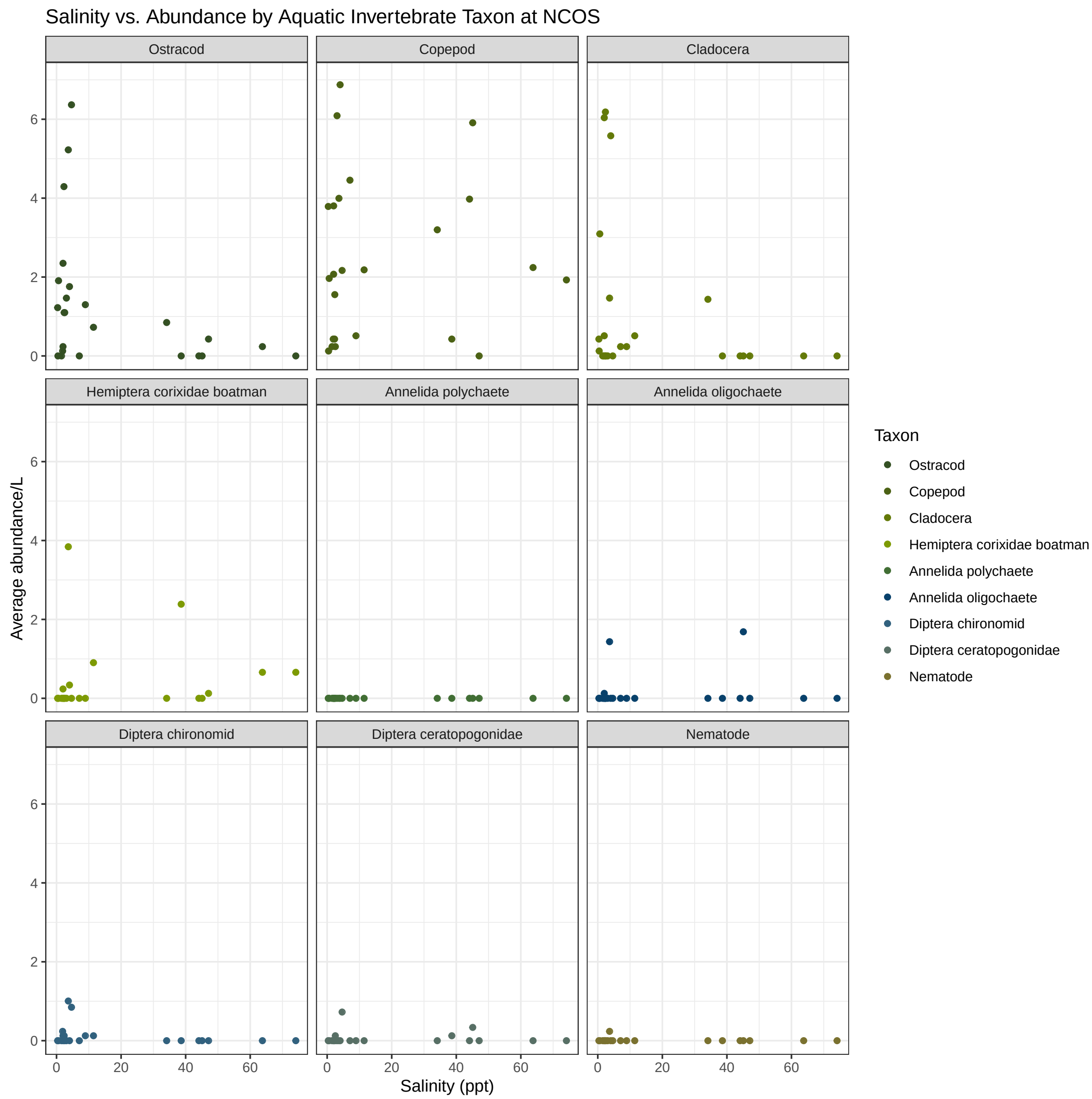
```
# A tibble: 10 x 3
   taxon                          med_sal log_ave_lit
   <fct>                            <dbl>       <dbl>
 1 Diptera chironomid                0.3        0
 2 Diptera chironomid                1.99       0.125
 3 Diptera chironomid               44.1        0
 4 Diptera chironomid                1.45       0
 5 Hemiptera corixidae boatman      NA          0
 6 Diptera ceratopogonidae           3.57       0
 7 Ostracod                         47          0.427
 8 Copepod                           4.57       2.17
 9 Nematode                         NA          0
10 Hemiptera corixidae boatman      NA          0.336
```

c. Code your figure

```r
# base layer
abu_sal_plot <- ggplot(data = abundance_salinity,
                   # x-axis: salinity
                   # y-axis: mean abundance/L
                   mapping = aes(x = med_sal,
                                 y = log_ave_lit,
                                 color = taxon)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
               fun = "median",
               size = 1) +
  facet_wrap(~ taxon) +
  # re labelling x- and y-axis, adding title
  labs(x = "Salinity (ppt)",
       y = "Average abundance/L",
```

```
       color = "Taxon",
       title = "Salinity vs. Abundance by Aquatic Invertebrate Taxon at
        ↪  NCOS") +
  # change color palette, indicate 9 colors for 9 taxon groups
  scale_color_manual(values = natparks.pals("Everglades", n = 9)) +
  # different theme
  theme_bw()

abu_sal_plot
```



d. Interpret your figure

Ostracods and Caldocera exist in the highest average abundance at lower salinities, between 0 and 10 ppt. Annelida polychaete, Annelida oligochaete, Diptera chironomid, Diptera ceratopogonidae and Nematodes all exist in low average abundance, typically less than 1/L, across the entire salinity range, 0-80 ppt. I could determine both of these patterns by observing where the highest concentration of data points lies along both the y and y axes. Copepods and

Hemiptera corixidae boatman are a bit more variable; there are data points high and low along both the average abundance and salinity axes.